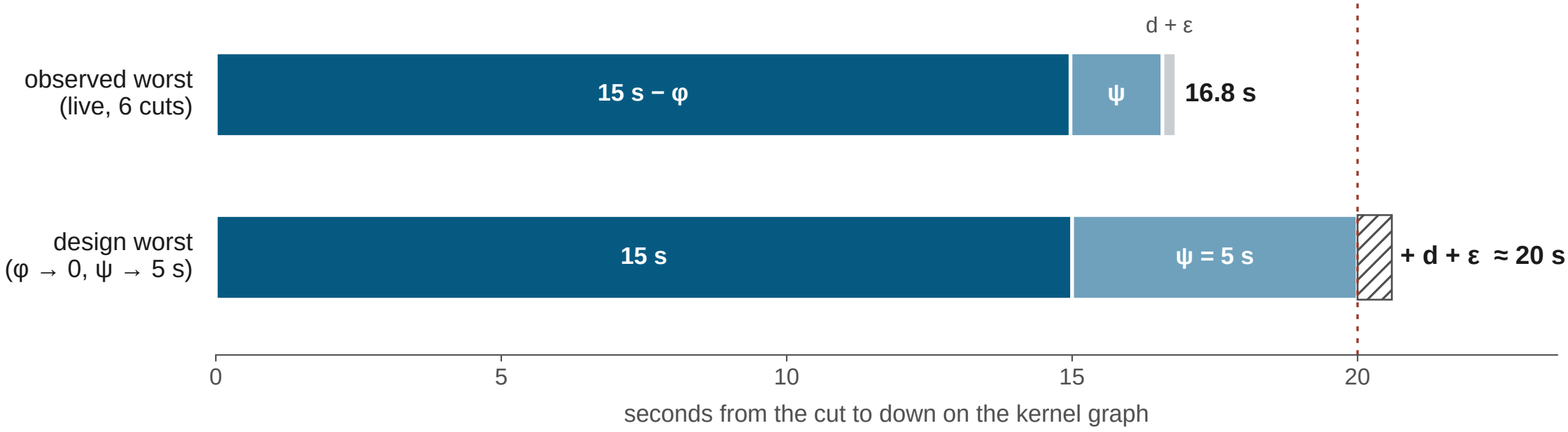The detection bound, piece by piece